PRD FOR CROPWATCH

Product Requirements Document · Team 8 · 10th Cohort · Version 1.0

PRODUCT OVERVIEW

Executive Summary

CropWatch is a mobile application that enables farmers to identify crop diseases and nutrient deficiencies using leaf photos and simple descriptions. Powered by AI, it provides real-time diagnoses and practical, locally available treatment recommendations.

By addressing the challenge of late disease detection, CropWatch helps farmers act early, reduce crop losses, and improve yields. The platform bridges the gap between agricultural expertise and rural communities, delivering accessible and affordable support to enhance productivity and food security.

PROBLEM STATEMENT

A farmer notices brown spots on their tomato leaves but cannot determine if it is blight, a leaf miner, or a nutrient deficiency. They often lack immediate access to expert advice, as the nearest agricultural extension office may be as far as 40 km away. By the time they can receive a diagnosis, the disease has often spread to neighboring plants, whereas early identification could have saved the entire harvest. Key problems include:

- No accessible real-time disease diagnosis tool for rural farmers
- Agricultural extension officers are too few and too far away
- Delayed diagnosis leads to crop loss and reduced income
- Treatment advice often recommends imported chemicals farmers cannot find or afford
- No centralized record of recurring disease problems on a farm
- Language barriers make existing online resources inaccessible

- Recommendations online are often:
  - Too generic
  - Not locally available
  - Too expensive

PRODUCT VISION AND GOAL

CropWatch is a mobile application that empowers farmers to diagnose crop diseases early using a photo of an affected leaf and a short text description of the symptoms. The AI engine analyzes the input and returns the most likely diagnoses with confidence levels, treatment recommendations using locally available solutions, and prevention tips for future seasons.

- Enable early crop disease detection through AI-powered mobile diagnosis
- Provide treatment recommendations using affordable, locally available solutions
- Support major Nigerian crops including tomato, cassava, maize, yam, pepper, rice, cowpea and cocoa
- Make the app accessible in both English and Pidgin English
- Give farmers a history log to track and monitor recurring problems on their farm
- Reduce post-harvest loss by putting expert-level knowledge in every farmer's pocket

TARGET USERS

User Type	Description	Primary Need
Smallholder Farmer	Rural farmers growing food crops in Nigeria and West Africa	Fast, reliable crop disease diagnosis on their phone
Commercial Farmer	Medium to large scale farm operators managing multiple crops	History tracking and recurring problem monitoring
Agricultural Student	Students studying agriculture or agronomy	Learning tool for identifying and understanding crop diseases
Extension Officer	Government agricultural extension workers in the field	A quick second-opinion tool during farm visits

KEY USE CASES

- **Use Case 1 — Immediate Disease Diagnosis:** A farmer in a remote area notices brown spots with yellow edges on tomato leaves. Instead of traveling 40 km to an office, they snap a photo and provide a brief text description in Pidgin English. Within seconds, the app identifies the likely disease and provides a confidence indicator, allowing the farmer to act before it spreads.

- **Use Case 2 — Localized Treatment Implementation:** Once a disease like leaf miner or blight is identified, the app suggests using wood ash or neem oil — remedies the farmer can find locally. This ensures the farmer is not forced to seek out expensive, imported chemicals they cannot find or afford.

- **Use Case 3 — Seasonal Prevention & Tracking:** After successfully treating a crop, the farmer reviews their History Log to see recurring problems from previous diagnoses. The app provides specific Prevention Tips for the next planting cycle, such as crop rotation strategies, helping the farmer move from reactive treatment to proactive farm management.

These use cases illustrate how CropWatch serves anyone who needs to quickly turn a leaf photo and symptom description into actionable agricultural advice. The platform is broad enough for various crops (tomato, pepper, maize, cassava, yam, rice, cowpea, cocoa) but focused on leveraging AI to boost harvest yields through early, affordable intervention.

MVP FEATURES

The following features define the MVP (Minimum Viable Product) for CropWatch. Each feature is described with its purpose and expected user outcome.

- **Photo Upload + Symptom Description:** The farmer takes or uploads a photo of the affected leaf and adds a short text description of what they observe (e.g. 'brown spots with yellow edges on tomato leaf'). Both inputs are sent to the AI for analysis.
- **AI-Powered Diagnosis:** The AI analyzes the combined photo context and text description to return the top 2–3 most likely causes of the problem, each with a confidence indicator (High, Medium, or Low). This helps the farmer understand how certain the diagnosis is.

- **Severity Indicator:** Each diagnosis includes a severity level: Mild (monitor the plant), Moderate (treat now), or Severe (remove affected plants immediately to prevent spread). This helps the farmer prioritize action.
- **Treatment Recommendations:** The app provides treatment advice that prioritizes locally available and affordable solutions first — neem oil, wood ash, crop rotation, companion planting — before suggesting commercial pesticides. This is critical for rural farmers with limited access to agro-chemical shops.
- **Prevention Tips:** After every diagnosis, the app provides tips on what to do next season to prevent the same problem from recurring. This builds long-term farming knowledge within the app.
- **Crop Disease Library:** A searchable reference library of common diseases and pests for the major Nigerian crops: tomato, pepper, maize, cassava, yam, rice, cowpea, and cocoa. Farmers can browse even without taking a photo.
- **Diagnosis History Log:** Every diagnosis is saved automatically to the farmer's history so they can track recurring problems, review past treatment advice, and monitor the health of their crops over time.
- **Language Toggle:** All content — diagnoses, treatment advice, and prevention tips — is available in both Standard English and Nigerian Pidgin English, making the app accessible to farmers with varying literacy levels.

FUTURE FEATURES

To evolve CropWatch from a diagnostic tool into a comprehensive agricultural ecosystem, the following roadmap outlines the planned expansion of its capabilities:

Phase 1: Enhanced Intelligence & Offline Support

- **Offline Mode for Remote Areas:** Implement a lightweight, on-device AI model or a caching system to allow farmers to capture photos and receive basic guidance even when they are 40 km away from stable internet.
- **Automated Image Recognition:** Move beyond text-based analysis to a fully integrated computer vision model that identifies diseases directly from the leaf photo without requiring a manual symptom description.
- **Audio Input (Voice-to-Text):** Add a voice note feature allowing farmers to describe symptoms verbally in English or Pidgin, which the AI then converts to text for analysis.

Phase 2: Community & Ecosystem Integration

- **Expert Consultation Bridge:** Create a 'Call an Expert' feature that allows farmers to share their diagnosis history and photos directly with a human agricultural extension officer for complex cases.
- **Localized Marketplace Integration:** Partner with local vendors to show farmers exactly where they can purchase the recommended 'locally available' treatments, such as neem oil or specific organic fertilizers, in their nearest town.
- **Community Disease Heatmap:** Use anonymized data from all users to create a real-time map showing disease outbreaks (e.g., a 'Blight Alert' for a specific region), helping neighboring farmers take preventive action.

Phase 3: Advanced Farm Management

- **Weather & Soil Integration:** Incorporate local weather forecasts and soil data to provide even more precise treatment and planting recommendations.
- **Predictive Yield Analysis:** Use the history of a farmer's diagnoses and treatments to estimate potential harvest yields and suggest improvements for the following season.
- **Expanded Crop Support:** Extend the database beyond the initial eight major Nigerian crops to include a wider variety of regional fruits, vegetables, and cash crops.

PAID VS FREE VERSION

CropWatch is designed to be accessible to every farmer, regardless of their financial standing. To ensure sustainability while fulfilling our mission of improving food security, the platform operates on a Freemium model. The Free version addresses the immediate '40 km gap' by providing essential diagnostics, while the Paid version offers advanced management tools for commercial success.

Feature Category	Free Version	Paid (Pro) Version
AI Diagnosis	Up to 5 leaf scans per day	Unlimited leaf scans and diagnoses
Symptom Input	Basic Identification + Confidence score	Severity Indicator + In-depth 'Why this happened' analysis
Treatment Guidance	Locally available/Organic solutions (Neem, ash, etc.)	Full access to the Industrial/Chemical treatment database
History & Records	Last 10 diagnoses saved in History Log	Unlimited History + Cloud Backup of all farm records

Offline Access	Basic symptom library (text-only)	Full Offline Mode (AI model runs without data/internet)
Reporting	View results on screen	Exportable PDF Reports for bank loans or insurance claims
Support	Community Forum + AI Bot	1-on-1 Chat/Video Call with Agricultural Extension Officers
Alerts	Standard regional outbreak notifications	Hyper-Local Alerts (Notifications for pests within a 5km radius)

The Free Version is built to be a life-saving tool for the everyday smallholder, focusing on immediate identification and affordable local remedies. The Paid Version ($10/month) is a professional-grade tool for commercial farmers, cooperatives, and extension officers who require unlimited data, offline functionality, and direct access to human experts.

DETAILED USER FLOW

Here is the end-to-end journey for a CropWatch user, from initial onboarding to receiving an AI-driven diagnosis and treatment plan.

1. Onboarding & Language Selection

- **Initial Launch:** The user opens the app and is greeted with a welcome screen.
- **Language Toggle:** Before proceeding, the user chooses between English and Pidgin English. The entire UI adjusts based on this selection to ensure clarity for rural farmers.
- **Account Creation (Optional/Pro):** The user can skip to immediate diagnosis or sign up/log in to save their history.
- **Dashboard:** A simple interface showing 'New Scan,' 'History,' and 'Prevention Tips.'

2. Diagnosis Request (The Input)

- **Start Scan:** The user taps the 'Identify Issue' button.
- **Image Capture:** The app opens the camera. The user snaps a clear photo of the affected crop leaf.
- **Symptom Description:** The user is prompted to describe what they see (e.g., 'The leaves are turning yellow and curling up').
- **Crop Selection:** The user selects the crop type from a list (e.g., Tomato, Cassava, Maize).

3. AI Analysis & Processing

- **Data Transmission:** The photo and text are sent to the AI engine (DeepSeek).
- **Processing:** The AI analyzes the visual patterns in the photo against the text description and the database of Nigerian crop diseases.
- **Retrieval:** The system pulls matching treatment data from the verified agricultural database (IITA/FAO).

4. Results & Severity Assessment (The Output)

- **Diagnosis Display:** The user receives the most likely identification (e.g., 'Early Blight') with a confidence percentage.
- **Severity Indicator:** The result is color-coded: Green (Mild) — Monitor the plant. Yellow (Moderate) — Treat now to prevent spread. Red (Severe) — Affected plants may need to be removed immediately.
- **Local Treatment Plan:** The app lists 2–3 affordable, locally available remedies (e.g., 'Mix wood ash with water' or 'Apply neem oil extract').

5. Follow-up & History

- **Save to History:** The diagnosis is automatically saved to the user's History Log for future reference.
- **Prevention Tips:** Based on the diagnosis, the app suggests steps for the next planting season to avoid a recurrence.
- **Expert Escalation (Paid Tier):** If the user is on the Pro plan and the case is 'Severe,' they can tap a button to send the report directly to a human extension officer for a 1-on-1 consultation.

TECHNICAL REQUIREMENTS

To ensure CropWatch remains functional and accessible for farmers in remote areas with varying internet stability, the following technical stack and requirements have been defined.

1. Frontend & Mobile Interface

- **Framework:** React Native or Flutter to ensure a smooth, native-like experience on both Android and iOS devices, with a focus on low-end Android phones common in rural areas.
- **Language Support:** Implementation of a global state management system (like Redux or Context API) to handle the real-time toggle between English and Pidgin English.

- **Image Processing:** Client-side image compression to ensure leaf photos can be uploaded quickly even on 2G or 3G networks.

2. Backend & API Layer

- **Server-Side Logic:** Node.js with Express to handle API requests between the mobile app, the AI engine, and the database.
- **AI Integration:** DeepSeek API for natural language processing of symptom descriptions and contextual diagnosis.
- **Computer Vision:** Integration with TensorFlow Lite for on-device image classification to reduce latency (Future).
- **Database:** Supabase (PostgreSQL) for managing user profiles, diagnosis history, and the library of Nigerian crop data.

3. Data Sourcing & Management

- **Agricultural Knowledge Base:** A structured database containing verified symptoms and treatments for the 8 core Nigerian crops, sourced from IITA (International Institute of Tropical Agriculture) and FAO open-access data.
- **Storage:** Supabase Storage or AWS S3 for secure hosting of user-uploaded leaf images.

4. Connectivity & Offline Functionality

- **Service Worker / Local Caching:** Use of SQLite for local storage, allowing farmers to access their History Log and Prevention Tips without an active internet connection.
- **Queue Management:** A 'Sync-when-Online' feature that allows users to capture a photo and description in the field (offline); the app automatically uploads and processes the diagnosis once a signal is detected.

5. Security & Privacy

- **Authentication:** Supabase Auth for secure user login via phone number or email.
- **Data Encryption:** SSL/TLS encryption for all data in transit between the mobile device and the server to protect farmer data and location privacy.

SUCCESS METRICS

To gauge CropWatch's performance and success, we will track a variety of key metrics. These metrics help us understand user adoption, engagement, the value delivered to customers, and overall business health. Below are the crucial metrics categorized by aspect:

1. User Adoption & Engagement

- **Total Active Farmers:** Number of unique farmers using the app to scan crops.
- **Diagnosis Volume:** Total leaf scans processed by the AI engine monthly.
- **Retention Rate:** Percentage of farmers who return for a second diagnosis within a planting season.

2. Revenue & Conversion (Paid Tier Success)

- **Free-to-Paid Conversion Rate:** Percentage of users who upgrade to the $10/month Pro Plan.
- **Monthly Recurring Revenue (MRR):** Total predictable revenue from individual and cooperative subscriptions.
- **Churn Rate:** Percentage of paid subscribers who cancel their subscription.

3. User Satisfaction & Trust

- **CSAT (Customer Satisfaction Score):** A simple 'Was this diagnosis helpful?' (Yes/No) or a 1–5 star rating after every treatment recommendation.
- **Trust & Accuracy Rating:** Percentage of users who manually confirm that the AI's identified disease matched the physical symptoms they saw.
- **App Store/Play Store Rating:** Monitoring public reviews to identify pain points in the user experience, especially regarding the Pidgin English toggle.
- **Referral Rate:** Number of new users who joined via a 'Share with a Fellow Farmer' link, indicating high community trust.

4. Diagnostic Accuracy & Efficacy

- **AI Confidence Levels:** Average confidence score provided by the AI (targeting >85%).
- **Resolution Rate:** Cases where a user marks a crop as 'Recovered' or 'Healthy' after following the suggested plan.

5. Impact on Harvest & Livelihood

- **Early Detection Rate:** Ratio of 'Mild' vs. 'Severe' diagnoses; success means catching more 'Mild' cases early.
- **Treatment Cost Savings:** Estimated savings by using local remedies versus expensive imported chemicals.

6. Technical Performance

- **Processing Latency:** Time from photo upload to diagnosis (targeting <30 seconds).
- **Offline Sync Success:** Reliability of the 'Sync-when-Online' feature in remote field conditions.

RISKS AND LIMITATIONS

While CropWatch provides a powerful AI-driven solution for rural farmers, there are several technical, environmental, and operational risks that could impact the effectiveness of the platform.

1. Accuracy of AI Diagnosis

- **Visual Mimicry:** Some crop diseases and nutrient deficiencies look remarkably similar in photographs (e.g., certain fungal spots vs. bacterial specks). There is a risk that the AI may misidentify a condition, leading to the application of the wrong treatment.
- **Image Quality Issues:** Photos taken in low light, blurry conditions, or with low-resolution cameras (common on budget devices) can significantly decrease the accuracy of the diagnostic engine.
- **Data Bias:** If the training data from sources like IITA or FAO does not sufficiently cover specific regional variations of pests in Nigeria, the AI may struggle to identify local 'edge cases.'

2. Connectivity and Technical Barriers

- **The '40 km Gap' Reality:** While the app aims to bridge the gap to extension offices, the very regions that need the app most often have the poorest 3G/4G coverage. If the 'Sync-when-Online' feature fails, farmers may not get the immediate help they need.
- **Device Limitations:** The app's performance may be hindered by older hardware, limited storage space on user devices, or fast battery drain — a major concern in areas with inconsistent electricity for charging.

3. Language and Cultural Nuance

- **Pidgin Translation Errors:** While the app includes a Pidgin English toggle, the nuances of regional dialects can vary. Misinterpreting a treatment instruction (e.g., the specific ratio for a wood ash mixture) could lead to crop damage rather than healing.
- **Trust and Adoption:** Farmers may be skeptical of 'machine advice' over traditional methods. A single high-profile failed diagnosis could damage the platform's reputation within a farming community.

4. Treatment and Resource Availability

- **Supply Chain Gaps:** The app may recommend 'locally available' remedies like neem oil or specific organic fertilizers, but if these are out of stock or overpriced in a specific village, the farmer remains unable to act.
- **Severity Misjudgment:** If a farmer misidentifies a 'Severe' case as 'Mild' due to user error during the input stage, they may delay removing a diseased plant, allowing the infection to spread to the rest of the farm.

COMPETITIVE LANDSCAPE AND ANALYSIS

CropWatch operates in the intersection of Agri-Tech and AI-driven diagnostics. While several global players exist, few focus specifically on the unique combination of Nigerian crop varieties, local language support (Pidgin), and low-cost 'backyard' treatment remedies.

- **Plantix:** A global leader in mobile crop diagnostics. It features a deep database of pests and diseases and a large community of users. Excellent for general identification but often lacks the hyper-local 'Pidgin English' accessibility and specifically curated 'local/organic' treatment alternatives tailored for the Nigerian market.
- **Agrio:** An AI-based precision agriculture solution that provides disease identification and satellite monitoring for large-scale farming. Highly sophisticated and great for commercial enterprises, but the interface can be overly technical for a small-scale rural farmer. It lacks the simplified '40 km gap' focus.
- **Leaf Doctor:** A specialized app that allows users to quantify the severity of plant diseases by selecting the diseased area on a photo. Provides great data on disease percentage, but does not offer the same level of automated treatment guidance or the local crop focus (like Cassava and Yam) found in CropWatch.
- **Government/NGO Extension Services:** Traditional human-led advisory services where officers visit farms or farmers visit central offices. High trust but extremely low scalability. The '40 km gap' makes this inaccessible for most rural farmers, leading to the 'Severe' crop loss cases CropWatch aims to prevent.

Competitive Summary Table

Platform	Strengths	Focus Area	Notes
Plantix	Large global database,	Global DIY Farmers	Broad but lacks local

	community forum		dialect/remedy focus
Agrio	Satellite monitoring, high-tech AI	Commercial & Enterprise	Can be expensive and technical for rural use
Leaf Doctor	Precise severity quantification	Research & Serious Hobbyists	Focused on data, not treatment or local crops
Extension Services	High human trust, expert verified	Regional Government	Inaccessible due to distance and low officer-to-farmer ratios
CropWatch	Local language, local remedies, AI speed	Rural Nigerian Smallholders	Bridges the 40 km gap with Pidgin and affordable solutions